Moumita Asad

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Education

Ph.D. in Software Engineering 2023 - 2029 (Expected)

University of California, Irvine, USA, CGPA: 4.00/4.00

Master of Science in Software Engineering (MSSE) 2019 - 2021

Institute of Information Technology, University of Dhaka, Bangladesh, CGPA: 3.94/4.00

Bachelor of Science in Software Engineering (BSSE) 2015 - 2018

Institute of Information Technology, University of Dhaka, Bangladesh, CGPA: 3.98/4.00

Notable Research Projects

GenLoc: Semantic Retrieval + LLM-Driven Code Exploration Jan 2024 - Jan 2026

- Developed GenLoc, combining embedding-based semantic retrieval with ReAct-style LLM-driven exploration and five code inspection tools to iteratively identify buggy source files.
- Evaluated across three Java and Python benchmarks, improving Accuracy@1 by **at least 63%** over traditional/deep learning methods and **25%** over recent LLM-based IRBL approaches.

Impact of Mislabeled Issues on Bug Localization Apr 2024 - Present

- Developed GenTag, an LLM-based issue classifier (weighted F1 = 0.90), revealing an average mislabeling rate of **35%** across five benchmarks covering **55K+** issue reports.
- Demonstrated that mislabeled issues inflate bug localization performance: MRR of six traditional, DL, and LLM-based techniques drops by up to **22%** on cleaned data.

Similarity-Based Automated Program Repair Jan 2019 - Dec 2020

- Developed patch prioritization techniques combining syntactic and semantic code similarity, reducing the search space needed to identify correct patches by **96%** across 246 bugs.
- Extended the techniques to two automated repair approaches for expression-level bugs, correctly repairing 22 and 21 bugs with overall precisions of 65% and 62%, respectively, across Defects4J and QuixBugs.

Notable Publications

Visit more on Google Scholar

- **Moumita Asad**, Rafed Muhammad Yasir and Sam Malek. "**Towards Explorative IRBL: Combining Semantic Retrieval with LLM-driven Iterative Code Exploration**". *ISSTA* (2026), pp. 1376-1400.
- Tasmia Zerin, **Moumita Asad**, B M Mainul Hossain and Kazi Sakib. "**Repairing Responsive Layout Failures Using Retrieval Augmented Generation**". *ICSME* (2025), pp. 845-850.
- **Moumita Asad**, Kishan Kumar Ganguly and Kazi Sakib. "**Combined Similarity Based Automated Program Repair Approaches for Expression Level Bugs**". *Communications in Computer and Information Science*, vol 1375. Springer, Cham (2021), pp. 311–335.
- **Moumita Asad**, Rafed Muhammad Yasir, Shihab Shahriar, Nadia Nahar and Md Nurul Ahad Tawhid. "**Analyzing Program Comprehensibility of Go Projects**". *SEKE* (2021), pp. 255-260.
- Toukir Ahammed, **Moumita Asad** and Kazi Sakib. "**Understanding the Relationship between Missing Link Community Smell and Fix-inducing Changes**". *ENASE* (2021), pp. 469-475.
- **Moumita Asad**, Kishan Kumar Ganguly and Kazi Sakib. "**Impact of Similarity on Repairing Small Programs: A Case Study on QuixBugs Benchmark**". *ICSE Workshops* (2020), pp. 21-22.
- Toukir Ahammed, **Moumita Asad** and Kazi Sakib. "**Understanding the Involvement of Developers in Missing Link Community Smell: An Exploratory Study on Apache Projects**". *QuASoQ@APSEC*(2020), pp. 64-70.
- **Moumita Asad**, Kishan Kumar Ganguly and Kazi Sakib. "**Impact of Combining Syntactic and Semantic Similarities on Patch Prioritization**". *ENASE* (2020), pp. 170-180.
- **Moumita Asad**, Kishan Kumar Ganguly and Kazi Sakib. "**Impact Analysis of Syntactic and Semantic Similarities on Patch Prioritization in Automated Program Repair**". *ICSME* (2019), pp. 328-332.
- Rafed Muhammad Yasir, **Moumita Asad**, Asadullah Hill Galib, Kishan Kumar Ganguly and Md Saeed Siddik. "**Godexpo: An Automated God Structure Detection Tool for Golang**". *IWoR@ICSE* (2019), pp. 47-50.

Professional Experience

Graduate Student Researcher, University of California, Irvine, USA Summers 2024 - 2026

– Project: GenLoc: Semantic Retrieval + LLM-Driven Code Exploration; Impact of Mislabeled Issues on IRBL

Intern, FoxAI, Bangladesh Jan 2018 - Jun 2018

– Implemented the front end of a chat platform which lets users create chatbots for Messenger

Requirement Analyst, Glitra Bangladesh Mar 2017 - May 2017

– Gathered and analyzed software requirements from the Department of Military Lands and Cantonments, Bangladesh, which led to the development of an automated system that streamlined their operations

Requirement Analyst, Softcell Solution Ltd Oct 2016 - Nov 2016

– Gathered and analyzed requirements from the Legal Sanctuary Law Firm, Bangladesh, enabling development of a case management system with case tracking, document search, billing, HR and accounting workflows

Teaching Experience

Graduate Teaching Assistant, University of California, Irvine, USA Sep 2023 - Present

– Courses: Software Testing, Distributed Software Architecture, Networking

Lecturer, University of Dhaka (On Study Leave) Feb 2022 - Present

– Courses: Software Requirement Analysis, Software Maintenance, Web Technology

– Supervising undergraduate students for their capstone projects

Lecturer, Independent University, Bangladesh Feb 2021 - Jan 2022

– Courses: Computer Fundamentals and Data Structure

– Faculty Advisor of the IUB ACM Student Chapter

Contractual Lecturer, BRAC University Oct 2020 - Jan 2021

– Courses: Artificial Intelligence

Academic and Professional Service

- **Journal Reviewer:** Automated Software Engineering (2024), Computing (2024)
- **Conference Reviewer:** MSR Junior PC (2027), ICSE Shadow PC (2027), CHI (2027), ReCode@ICSE (2026), ICSA (2025), ICSA Artifact Evaluation Track (2024)
- **Community Service:** Judge at UCI Undergraduate Research Symposium, UCI Undergraduate Research Opportunities Program (UROP) (2025), Instructor at Code in Place, Stanford University (2021)

Notable Software Projects

GenArch – Generative AI-based Software Architecture Recovery: Developed an LLM-based tool to recover software architecture using source code summarization and embedding-based clustering (Python).
Show Me Related Work: Built a research paper citation graph web app using Python and Angular 4; ranked among the top six projects at SCORE ICSE 2018.
MalFinder – Cloud-based Malware Detection System: Developed a malware detection system combining static and dynamic analysis (Python Flask, Java Swing).

Virtual Trial Room: Developed an AR-based tool for real-time virtual clothing fitting (Python, OpenCV).

Honors and Awards

- **ACM SIGSOFT CAPS travel grant** for attending ISSTA - 2026
- **Chair's Award**, University of California, Irvine, in recognition of exceptional promise in research - 2023
- **IIT Academic Excellence Gold Medal Award** for obtaining the highest CGPA in MSSE - 2020
- **Fellowship from ICT Division**, Ministry of Posts, Telecommunications and Information Technology, Bangladesh. Dated 20.11.2019
- **IIT Academic Excellence Gold Medal Award** for obtaining the highest CGPA in BSSE - 2018
- **Talent pool scholarship and full tuition waiver** in MSSE for excellence in BSSE, University of Dhaka - 2018
- **Champion**, Technovation Hackathon, North South University - 2018